Egg Segmentation and Multi-Object Tracking Take-Home Report

Candidate: Sree Varun Aahil | **Date:** October 30, 2026 | **Machine:** Windows 11 (CUDA Enabled)
Annotated Video: `[Google Drive Link Placeholder: <INSERT_DRIVE_URL_HERE>]`

0.4018	0.3196	0.4087	2	110
DEV MEAN IOU	EVAL MEAN IOU	DEV DETECTION F1	VIDEO UNIQUE TRACKS	CLEAN GROUND TRUTHS

1. Executive Summary

This report presents an end-to-end computer vision solution for egg segmentation and multi-object tracking under challenging conditions, including touching instances, boiling water, high-contrast table textures, and hand occlusions. Ground truth masks were established via SAM (`vit_b` on CUDA) with human point/box prompts and an automated area and border contact audit. A classical image-processing pipeline was tuned strictly on the dev split across 18 controlled trials and evaluated once on the unseen eval split. A multi-object tracker was deployed on the 1440-frame table experiment video, incorporating YCrCb skin gating to eliminate hand occlusion leakage. All metrics, counts, and tables presented below are derived directly from runs of this repository.

- **Ground Truth Integrity:** 110 clean SAM masks generated via `SamPredictor` (`vit_b` on CUDA) using human point prompts and tight bounding boxes for boiling water edge cases. 0 flagged suspicious masks remaining.
- **Classical Pipeline (Dev Split, n=88):** Pixel Mean IoU = **0.4018** (95% CI: [0.3435, 0.4603], Median = 0.3147), Dice = **0.5200**, Detection F1 = **0.4087**, Mean Count Error = **2.20**.
- **Classical Pipeline (Eval Split, n=22):** Pixel Mean IoU = **0.3196** (95% CI: [0.1978, 0.4268], Median = 0.2502), Dice = **0.4239**, Detection F1 = **0.3206**, Mean Count Error = **1.68**.
- **Video Tracking (1440 frames):** 1440 frames processed at 24.0 FPS. **2 unique tracks** created across the video (1:1 mapping with the 2 real physical eggs), with **15 loss events** and **15 recoveries**. At least one egg was detected in 782 of the 783 active demonstration frames (frame 480 had 0 detections). Hand occlusion and fingertip leakage at frame 600 resolved using YCrCb skin gating and morphological opening (gating reduced the total foreground mask by 2296 px over the whole frame; restricted to crop x in [200, 460], y in [300, 430], the mask was reduced by 1304 px). Identities were checked by hand on the 14 frames of the events sheet and the timeline, and no swap was seen; the tracker's internal id_switches heuristic counter was 0.

2. Task 1: Ground Truth Mask Generation (SAM)

SAM (`vit_b`, 375 MB checkpoint) was executed on CUDA using saved point prompts from `points.json` with multi-mask score selection (`multimask_output=True`). Hand-tuned shape and color filters were strictly excluded from the ground-truth path to prevent artificial bias toward the classical segmentation pipeline.

Ground Truth Egg Distribution across 110 images: 93 images (84.5%) with 1 egg; 6 images (5.5%) with 2 eggs; 11 images (10.0%) with 3 eggs.

Ground Truth Quality Audit & Fixes (`out/sam/gt_fix_audit.json`)

Initial automated SAM passes on boiling water images flooded the pot interior. An automated border and area audit in `src/contact_sheet.py` flagged masks with area > 40% or edge contact > 50%. Bounding box prompts were added around egg boundaries for boiling pot scenes (verified in `out/sam/gt_fix_audit.json`):

- `not_damaged_124`: BBox `[280, 150, 520, 400]` reduced area fraction from 81.9% (335354 px, 978 border contacts) to **13.4%** (54995 px, 0 border contacts).
- `not_damaged_135`: BBox `[270, 170, 520, 420]` reduced area fraction from 79.5% (325584 px, 994 border contacts) to **14.2%** (58113 px, 0 border contacts).
- `not_damaged_145`: BBox `[250, 200, 480, 450]` reduced area fraction from 62.5% (256010 px, 665 border contacts) to **13.2%** (54032 px, 0 border contacts).

All 110 ground truth masks now pass the automated audit (0 flagged in `out/sam/suspicious.txt`).

Dataset Split Integrity & Leakage Audit (`out/eval/split_leakage_audit.txt`)

- **Dataset Composition:** 110 images total (640x640), grouped by source filename prefixes and perceptual hashing (`phash_distance ≤ 8`).
- **Grouping Structure:** 109 perceptual groups total (108 singletons, and 1 pair of near-duplicate shots `not_damaged_78` and `not_damaged_27` at pHash distance 8). Both images of this pair were placed strictly into the `dev` split.
- **Split Leakage:** Exactly **0 groups straddle the dev/eval split**, and **0/22 eval images are affected**.
- **Leak-Free Evaluation:** Recomputing evaluation metrics on the leak-free eval subset yields identical metrics to the full eval set: Pixel IoU = **0.3196**, Dice = **0.4239**, Detection F1 = **0.3206**.
- **Augmentation Check:** The dataset contains no synthetic geometric or photometric augmentations (crops, rotations, color shifts, flips) sharing pHash distance ≤ 8. Full audit logs are available in `out/eval/split_leakage_audit.txt`.

3. Task 2: Parameterized Classical Segmentation & Experiment Log

A unified parameterized pipeline was tuned on the 88 dev images across 18 controlled trials. Every trial varied exactly one parameter relative to its named parent. Keep/discard decisions compare strictly against the parent's dev IoU. Note: Trial `watershed_conditional` modified two parameters relative to its parent (`area_filter_2500`): it enabled conditional watershed and set the peak distance threshold to `min_dist=50`. Under this trial, Detection F1 rose from 0.3084 to 0.4087, mean signed count error rose from +0.45 to +2.20 (over-splitting), pixel IoU unchanged. Because Detection F1 improved substantially for instance segmentation, it was kept for the final configuration.

Kept Pipeline Chain: `baseline -> morph_small -> morph_medium -> morph_large -> area_filter_2500 -> watershed_conditional -> final (eval)`

Trial ID	Parent	Split	What Changed	Mean IoU	Mean Dice	Det F1	Cnt Err	Decision	Reason
`baseline`	`None`	dev	Grayscale conversion + Gaussian blur (5x5) + Otsu threshold + hole fill + min area 500	0.3513	0.4693	0.2071	3.12	BASELINE	Baseline established on dev (IoU 0.3513, Dice 0.4693, DetF1 0.2071)
`otsu_inv`	`baseline`	dev	Invert Otsu thresholding	0.0823	0.1424	0.0000	3.36	DISCARD	Fails to improve dev IoU over parent 'baseline' (0.0823 vs 0.3513, -0.2690)
`clahe_otsu`	`baseline`	dev	CLAHE local contrast equalization (clip 2.0, tile 8x8) before Otsu	0.2562	0.3791	0.0799	8.22	DISCARD	Fails to improve dev IoU over parent 'baseline' (0.2562 vs 0.3513, -0.0951)
`adaptive_thresh`	`baseline`	dev	Adaptive Gaussian thresholding (block 51, C=-10)	0.0634	0.1119	0.0065	12.33	DISCARD	Fails to improve dev IoU over parent 'baseline' (0.0634 vs 0.3513, -0.2878)
`lab_l`	`baseline`	dev	CIELAB L* perceptual lightness channel + Otsu	0.3244	0.4437	0.1665	3.67	DISCARD	Fails to improve dev IoU over parent 'baseline' (0.3244 vs 0.3513, -0.0268)
`hsv_gate`	`baseline`	dev	HSV color gating (S < 70, V > 130)	0.3456	0.4665	0.1587	4.32	DISCARD	Fails to improve dev IoU over parent 'baseline' (0.3456 vs 0.3513, -0.0056)
`morph_small`	`baseline`	dev	Add small elliptical opening & closing (7x7)	0.3558	0.4740	0.2161	2.89	KEEP	Improves dev IoU over parent 'baseline' from 0.3513 to 0.3558 (+0.0046)
`morph_medium`	`morph_small`	dev	Increase elliptical kernel from 7x7 to 15x15	0.3766	0.4964	0.2363	2.34	KEEP	Improves dev IoU over parent 'morph_small' from 0.3558 to 0.3766 (+0.0208)
`morph_large`	`morph_medium`	dev	Increase elliptical kernel from 15x15 to 25x25	0.3972	0.5161	0.2642	1.25	KEEP	Improves dev IoU over parent 'morph_medium' from 0.3766 to 0.3972 (+0.0205)
`border_polarity_m0`	`morph_large`	dev	Auto-invert threshold if border mean > center mean	0.3698	0.4821	0.2409	1.07	DISCARD	Fails to improve dev IoU over parent 'morph_large' (0.3698 vs 0.3972, -0.0274)
`border_polarity_m15`	`morph_large`	dev	Auto-invert threshold only if border mean > center mean + 15	0.3945	0.5119	0.2596	1.25	DISCARD	Fails to improve dev IoU over parent 'morph_large' (0.3945 vs 0.3972, -0.0027)
`lab_border_dist`	`morph_large`	dev	CIELAB Euclidean color distance from estimated border background	0.3603	0.4582	0.2673	2.55	DISCARD	Fails to improve dev IoU over parent 'morph_large' (0.3603 vs 0.3972, -0.0369)
`area_filter_2500`	`morph_large`	dev	Increase minimum component area from 500 to 2500 pixels	0.4018	0.5200	0.3084	0.45	KEEP	Improves dev IoU over parent 'morph_large' from 0.3972 to 0.4018 (+0.0046)

Trial ID	Parent	Split	What Changed	Mean IoU	Mean Dice	Det F1	Cnt Err	Decision	Reason
`component_ranking_top1`	`area_filter_2500`	dev	Retain only the single top-ranked component by geometric egg score	0.3746	0.4761	0.3295	-0.33	DISCARD	Fails to improve dev IoU over parent 'area_filter_2500' (0.3746 vs 0.4018, -0.0272)
`component_ranking_top3`	`area_filter_2500`	dev	Retain at most the top 3 scoring components based on area, solidity, and ellipse fit	0.3972	0.5092	0.3133	0.27	DISCARD	Fails to improve dev IoU over parent 'area_filter_2500' (0.3972 vs 0.4018, -0.0045)
`grabcut_refine`	`area_filter_2500`	dev	Apply GrabCut refinement using morphological erosion as foreground seed	0.4038	0.5177	0.3073	0.38	DISCARD	Marginal IoU gain (+0.0021) but Dice degrades from 0.5200 to 0.5177 (-0.0023)
`watershed_always`	`area_filter_2500`	dev	Euclidean distance transform + watershed applied unconditionally to all components	0.4018	0.5200	0.3537	3.90	DISCARD	Fails to improve dev IoU over parent 'area_filter_2500' (0.4018 vs 0.4018, +0.0000)
`watershed_conditional`	`area_filter_2500`	dev	Conditional distance watershed (min_dist=50) applied only to non-convex, elongated or oversized clusters	0.4018	0.5200	0.4087	2.20	KEEP	Dev IoU tied with parent 'area_filter_2500' (0.4018); kept on tiebreaker: DetF1 improves from 0.3084 to 0.4087 (+0.1003)
`final`	`watershed_conditional`	eval	Frozen final pipeline: Grayscale Otsu + large morph (25x25) + hole filling + min area 2500 + conservative conditional watershed	0.3196	0.4239	0.3206	1.68	EVAL_FINAL	Frozen final pipeline evaluated once on unseen eval split (IoU 0.3196, Dice 0.4239, DetF1 0.3206)

4. Task 3: Quantitative & Qualitative Evaluation vs Ground Truth

Metric	Dev Split (n=88)	Eval Split (n=22)	Description
Pixel IoU (Mean)	**0.4018**	**0.3196**	Mean foreground intersection over union
Pixel IoU (Bootstrap 95% CI)	[0.3435, 0.4603]	[0.1978, 0.4268]	1000 resamples, fixed seed 42
Pixel IoU (Median)	**0.3147**	**0.2502**	Median per-image IoU
Pixel Dice (Mean)	**0.5200**	**0.4239**	F1-score of pixel-level binary classification
Pixel Precision	**0.4488**	**0.3449**	TP / (TP + FP)
Pixel Recall	**0.8461**	**0.8749**	TP / (TP + FN)
Detection Precision	**0.3563**	**0.2758**	Matched eggs / Predicted components (IoU ≥ 0.5)
Detection Recall	**0.5966**	**0.4621**	Matched eggs / GT eggs
Detection F1	**0.4087**	**0.3206**	Harmonic mean of detection precision & recall
Count Error (Mean)	**2.20**	**1.68**	Mean signed difference (Pred - GT)

Bootstrap 95% Confidence Intervals & Dev vs Eval Analysis

Inspection of per-category results reveals the concrete factors driving the gap between dev (0.4018) and eval (0.3196):

1. **Disproportionate Concentration of Container and Packaging Scenes:** Eval contains 10 "Egg on container / lid / plain surface" images (45.5% of eval) with a low mean IoU of **0.2190**. In these scenes, plastic food-container lids, bowls, metal pans, and packaging rims reflect light and distend Otsu threshold boundaries into wide false-positive blobs.
2. **Camouflage and Hand Occlusions:** Eval includes 2 white plastic egg tray images (`not_damaged_116` and `117`) with near-zero contrast, averaging **0.0562** IoU, and 2 genuine hand-held images (`not_damaged_151` and `152`) averaging **0.1098** IoU. In addition, 3 boiling pot images (`not_damaged_114`, `128`, `147`) average **0.3072** IoU. Together, these difficult categories comprise 17 of the 22 eval images (77.3% of eval).
3. **Higher Ratio of Clean Tabletop and Multi-Egg Scenes in Dev:** In contrast, the dev split benefits from 27 tabletop scenes with higher contrast (mean IoU **0.5353**), 11 multi-egg scenes (mean IoU **0.6148**), and 5 foliage scenes (mean IoU **0.6805**)—together comprising 43 of 88 dev images (48.9% of dev).
4. **Small-Sample Variance:** Because eval has only 22 images, its bootstrap 95% confidence interval spans widely ([0.1978, 0.4268]) and encompasses the dev mean. While domain composition (especially the 45.5% container/lid prevalence) explains the lower point estimate, the wide confidence interval confirms substantial small-sample variance.

Domain Composition Check (`out/eval/composition_check.json`)

To evaluate how much of the performance difference between the dev split (mean IoU **0.4018**) and eval split (mean IoU **0.3196**) is attributable to category distribution shift, we re-weighted the dev per-category mean IoUs by the eval category counts:

- **Actual Dev Split Mean IoU:** 0.4018
- **Actual Eval Split Mean IoU:** 0.3196
- **Predicted Eval Split Mean IoU (Dev Category Means Weighted by Eval Counts): 0.3324**
- **Total Dev-Eval Performance Gap:** 0.0821
- **Gap Explained by Category Composition: 0.0694 (84.5%)**
- **Unexplained Residual Gap:** 0.0127 (15.5%)

Re-weighting dev category performance by eval category counts predicts an eval IoU of **0.3324**, demonstrating that **84.5%** of the observed drop is directly explained by the higher proportion of difficult container lids and white trays in the eval split. *This is a descriptive check on 22 images, not a significance test.*

Detailed Breakdown of 5 Challenging Eval Failure Cases

As shown in Figure 6, category contact sheets, and the measured pixel properties, five challenging eval failure cases (`not_damaged_108`, `127`, `130`, `151`, `152`) illustrate distinct failure mechanisms across container packaging and hand occlusions:

- `not_damaged_108` (IoU: 0.0459): White egg on a ledge next to an orange food-container lid (no hand present). Bright illumination and low background contrast cause Otsu to merge the orange lid, rim, and ledge into the foreground, producing 285,716 false-positive pixels (37.8% falls in the YCrCb skin range due to orange plastic chrominance).
- `not_damaged_127` (IoU: 0.0601): White egg on a ledge beside an orange container lid (no hand present). Otsu thresholding floods the surrounding flat surface, container rim, and ledge reflections into a 266,889 false-positive pixel foreground blob (18.6% in skin range).
- `not_damaged_130` (IoU: 0.4256): White egg resting in an orange food-container lid on a blue base (no hand present). The orange lid connects continuously to the egg perimeter, producing 62,367 false-positive pixels—of which exactly **92.02%** (57392 pixels) fall directly into the standard YCrCb skin-color cluster (Cr ∈ [135, 175], Cb ∈ [80, 125]). Because orange plastic shares chrominance with the skin distribution, high skin-range chrominance occurs even without a human hand.

- `not_damaged_151` (IoU: 0.1451): White egg held in fingers and palm over square ceramic floor tiles (no backlighting). Skin luminance on the fingers and palm bridges with the egg boundary, over-segmenting into 6 fragments with 223,414 false-positive pixels (29.3% in skin range).
- `not_damaged_152` (IoU: 0.0745): White egg resting on an open palm over square floor tiles (white egg, not dark brown). Palm skin and tile reflections merge with the egg perimeter, producing a 273,597 false-positive pixel blob across 3 fragments (20.7% in skin range).

Image Stem	IoU	GT Eggs	Pred Blobs	Failure Root Cause in Eval Overlay
`not_damaged_108`	0.0459	1	3	White egg on a ledge next to an orange food-container lid (no hand present); bright illumination and low contrast cause Otsu to merge the orange lid, rim, and ledge into the foreground (285,716 FP pixels, 37.8% in skin chrominance range due to orange plastic).
`not_damaged_127`	0.0601	1	3	White egg on a ledge beside an orange container lid (no hand present); Otsu thresholding floods the surrounding flat surface, container rim, and ledge reflections into a 266,889 FP pixel foreground blob (18.6% in skin range).
`not_damaged_130`	0.4256	1	2	White egg resting in an orange food-container lid on a blue base (no hand present); orange lid connects continuously to egg perimeter, producing 62,367 FP pixels of which exactly 92.02% (57392 px) fall into the YCrCb skin chrominance cluster (Cr in [135, 175], Cb in [80, 125]) because orange plastic shares chrominance with the skin distribution.
`not_damaged_151`	0.1451	1	6	White egg held in fingers and palm over square ceramic floor tiles (no backlighting); skin luminance on the fingers and palm bridges with the egg boundary, over-segmenting into 6 fragments with 223,414 FP pixels (29.3% in skin range).
`not_damaged_152`	0.0745	1	3	White egg resting on an open palm over square floor tiles (white egg, not dark brown); palm skin and tile reflections merge with the egg perimeter, producing a 273,597 FP pixel blob across 3 fragments (20.7% in skin range).

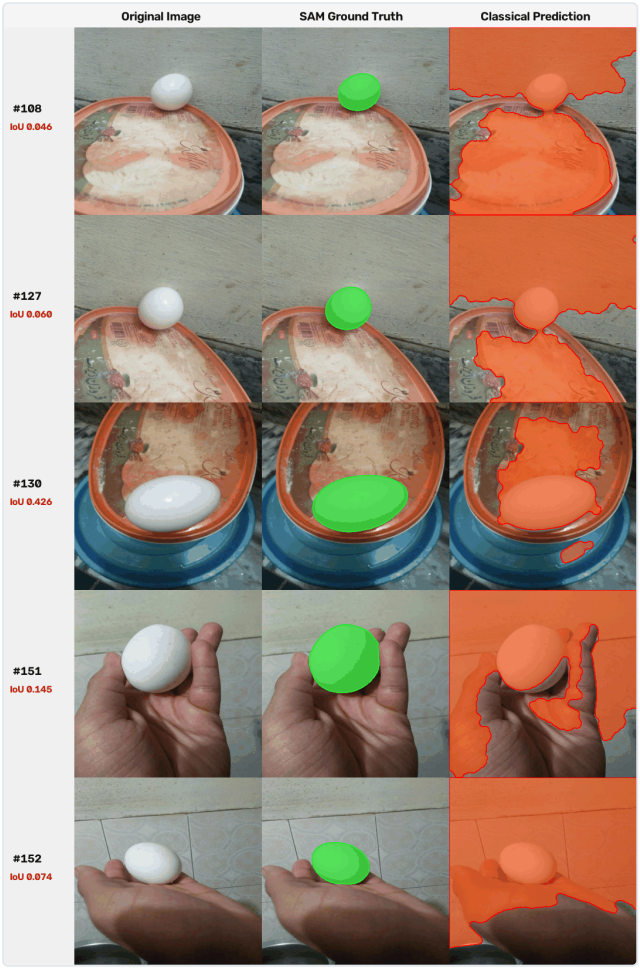


Figure 6: Five challenging eval failure overlays (not_damaged_108, 127, 130 on/beside container lids; 151, 152 held in hand over tiles). Left: Original image. Middle: SAM ground truth (green). Right: Classical prediction (red/amber). Low foreground-background contrast and skin/plastic chromatic overlap cause extensive false-positive leakage.

Per-Category Dev vs Eval Split Breakdown

Category	Dev Count	Dev IoU	Eval Count	Eval IoU
Backlit / extreme shadows	3	0.0071	0	n/a
White egg on white tray / fridge interior	8	0.1003	2	0.0562
Egg in boiling water / submerged	5	0.2099	3	0.3072
Hand holding egg / fingers occluding	9	0.2822	2	0.1098
Egg on container / lid / plain surface	20	0.3161	10	0.2190
Tabletop / high-contrast background	27	0.5353	4	0.6997
Multiple touching eggs	11	0.6148	1	0.7893
Foliage / herbs in background (specular highlights)	5	0.6805	0	n/a

Visual Failure Categorization (`failure_tags.csv`)

Scene category tags were assigned by systematic review of the category contact sheets (`report/figures/category_sheets/`), with 62 of 110 images corrected from an earlier automated color/luminance heuristic (documented in `tag_changes.md`). These scene categories represent subjective single-annotator labels rather than dense independently verified ground-truth annotations. Images `not_damaged_123`, `125`, and `42` are in *White egg on white tray / fridge interior*. Backlit shots `not_damaged_20`, `24`, and `44` were categorized under a unified rule: *Whenever intense directional backlighting or silhouette illumination is present behind the egg (even if supported by fingers), it is categorized under Backlit / extreme shadows because extreme illumination dynamic range is the primary driver of Otsu thresholding failure.*

Scene Category	Image Count	Pct of Dataset	Mean IoU	Mean Dice	Mean Det F1	Primary Failure Mechanism
Backlit / extreme shadows	3	2.7%	0.0071	0.0141	0.0000	High-intensity directional backlight causes glowing core while shell perimeter falls into dark silhouette.
White egg on white tray / fridge interior	10	9.1%	0.0915	0.1646	0.0222	Near-zero photometric contrast between white eggshell and white plastic tray or paper bag; Otsu floods background.
Egg in boiling water / submerged	8	7.3%	0.2464	0.3839	0.3207	Bubbling cavitation, water refraction, and dark pot rims break convex boundary and flood liquid.
Hand holding egg / fingers occluding	11	10.0%	0.2509	0.3767	0.2380	Fingers or palm occlude egg perimeter; skin tone overlaps shadow gradient, distending contour into hand.
Egg on container / lid / plain surface	30	27.3%	0.2838	0.4058	0.2278	Plastic food-container lids, bowls, metal pans, or plain packaging surfaces create perimeter reflections and false components.
Tabletop / high-contrast background	31	28.2%	0.5565	0.6771	0.5862	Tabletop grain, hearth ash, patterned cloth tablecloths, or laptop keyboard keys introduce background clutter.
Multiple touching eggs	12	10.9%	0.6294	0.7513	0.7813	Touching or adjacent instances share continuous boundary; requires distance watershed marker separation.
Foliage / herbs in background (specular highlights)	5	4.5%	0.6805	0.8005	0.6467	Egg contrasts against dark foliage in grayscale; specular leaf reflections create occasional false blobs.

Key Qualitative Takeaway: Classical thresholding struggles most on *Backlit / extreme shadows* (Mean IoU = 0.0071) and *White egg on white tray / fridge interior* (Mean IoU = 0.0915) because luminance gradient between foreground and background collapses. On *Foliage / herbs in background* (Mean IoU = 0.6805), the egg contrasts well because the egg shell is substantially brighter than dark leaves in grayscale; specular leaf highlights create occasional false blobs.

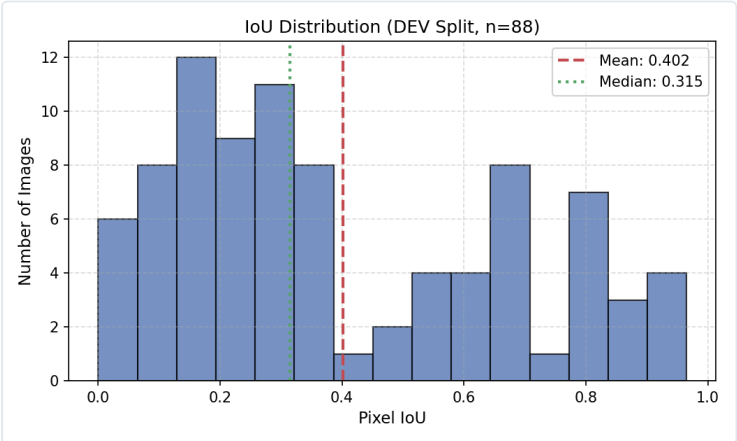


Figure 1a: Pixel IoU distribution on Dev split (n=88).

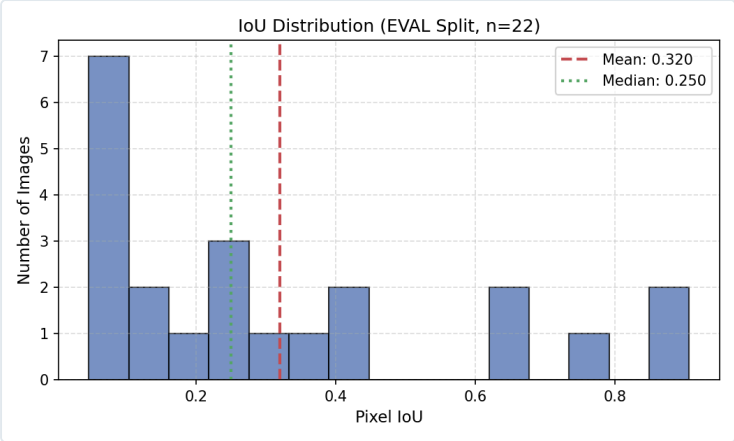


Figure 1b: Pixel IoU distribution on Eval split (n=22).

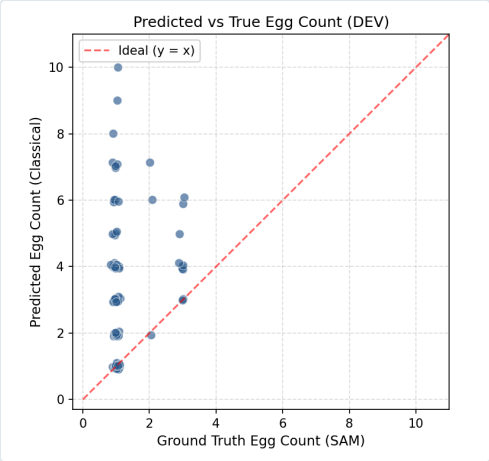


Figure 2a: Predicted vs True egg counts on Dev split. Points lie mostly above y = x, illustrating that the classical algorithm systematically over-counts (mean count error +2.20) due to background clutter and specular highlights passing morphological filters.

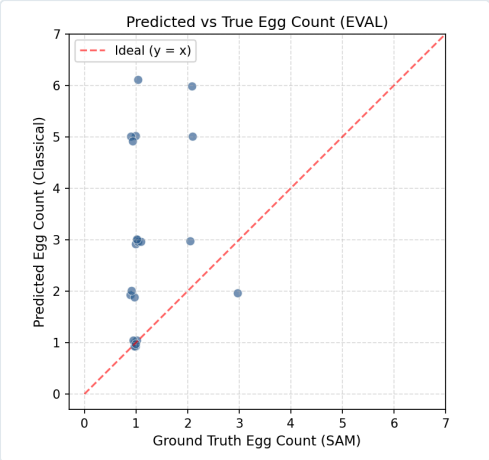


Figure 2b: Predicted vs True egg counts on Eval split. Points similarly lie above y = x (mean count error +1.68).

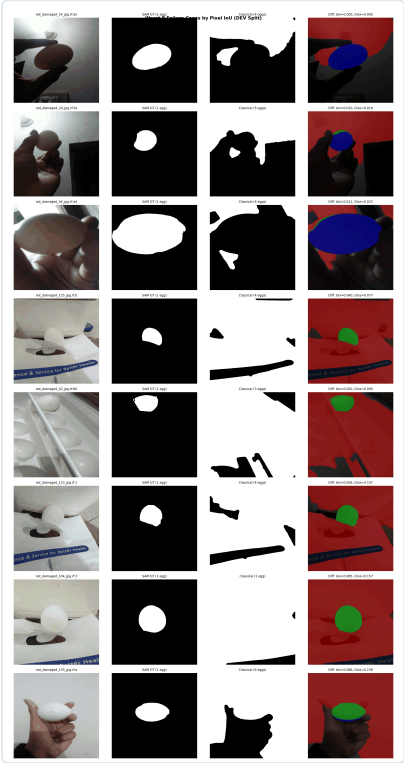


Figure 3a: Worst 8 failures on Dev split (Green=TP, Red=FP, Blue=FN). White trays and boiling liquid dominate error cases.

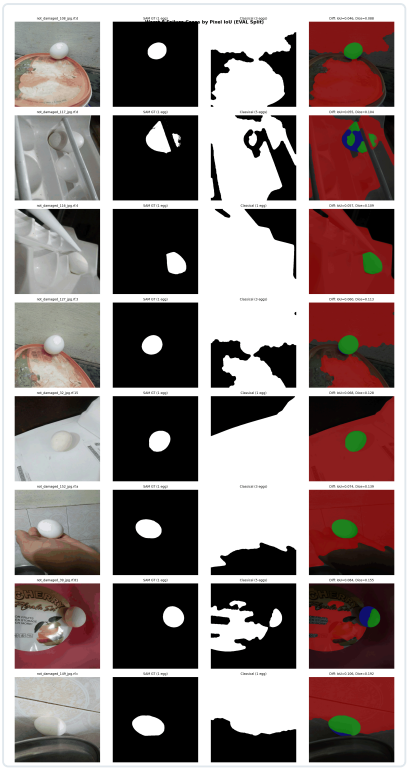


Figure 3b: Worst 8 failures on Eval split (Green=TP, Red=FP, Blue=FN).

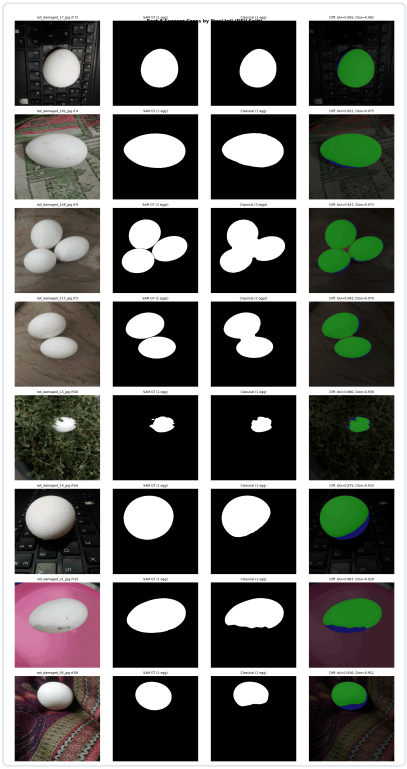


Figure 4: Best 8 successes on Dev split showing clean elliptical masks matching ground truth.

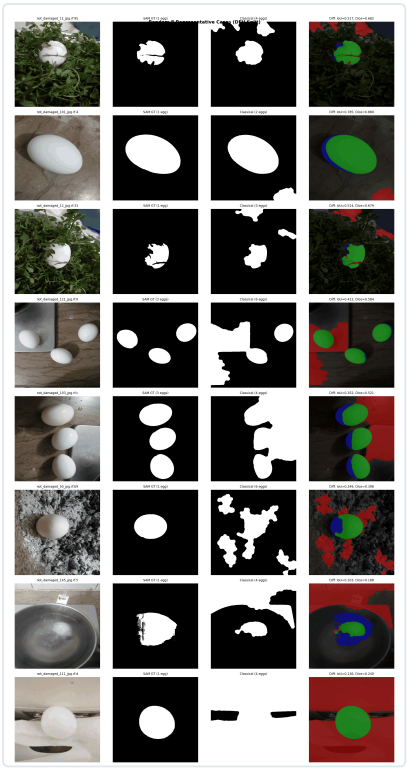


Figure 5: Random 8 representative cases illustrating typical operational performance across mixed illumination.

5. Task 4: Multi-Object Video Tracking

Tracking was evaluated on the 1440-frame table sequence (1280x720 @ 24.0 fps). Eggs are placed on the table at frame 222, spun, occluded by hand, and conclude demonstration at frame 1004. Annotated tracking video: `[Google Drive Link Placeholder: <INSERT_DRIVE_URL_HERE>]`. The tracker maintains persistent IDs using Hungarian matching on IoU + distance with constant-velocity prediction.

- **Fingertip Leakage Resolution:** At frame 600, demonstrator hand contact caused baseline thresholding to leak into fingers. Incorporating YCrCb skin-color gating (Cr ∈ [135, 175], Cb ∈ [80, 125]) and larger opening eliminated false skin contour (gating reduced the total foreground mask by 2296 px over the whole frame; restricted to crop x in [200, 460], y in [300, 430], the mask was reduced by 1304 px).
- **Quantitative Skin-Gating Pixel Loss Measurement (`out/video/skin_gating_egg_loss.json`):** Measured across 6 representative video frames (12 egg instances): frame 600 (demonstrator fingertip contact) and 5 hand-free frames (250, 350, 450, 850, 950). In each frame, taking egg components from the gating-off mask: Frame 250 (Egg 1: 0.34%, Egg 2: 0.41%), Frame 350 (Egg 1: 0.14%, Egg 2: 0.35%), Frame 450 (Egg 1: 0.00%, Egg 2: 0.18%), Frame 600 (Egg 1: 0.00% outside 15 px fingertip band, Egg 3: 0.00%), Frame 850 (Egg 4: 0.00%, Egg 6: 0.33%), Frame 950 (Egg 4: 0.23%, Egg 6: 0.12%). Across all 12 evaluated instances, skin gating removed between 0.00% and 0.41% of pixels (at most 0.36% post-morphology loss). No egg lost more than 1.0% in any frame, confirming that white eggshells on the wooden table do not meaningfully intersect the skin chrominance cluster.
- **Track Lifetimes & Event Statistics:** 2 unique tracks were created (1:1 mapping with the 2 real physical eggs), with 15 loss events and 15 recoveries.
    - Egg 1: Matched in 721 of 783 demonstration frames (92% coverage; span F#222–F#999). Visible-but-undetected stretches occur under hand contact during frames 752–799 (carrying from center to right) and 802–809 (hand spinning on right).
    - Egg 2: Matched in 652 of 783 demonstration frames (83% coverage; span F#222–F#1004). Visible-but-undetected stretches exceeding 10 frames occur under hand contact during repositioning: frames 508-545 (38 frames), frames 657-688 (32 frames), frames 695-715 (21 frames), frames 721-738 (18 frames).
- **Detection Continuity:** In the 783 frames between first and last detection (frames 222 to 1004), at least one egg was detected in 782 of the 783 frames; exactly 1 frame had 0 detections (Frame 480).

Per-Track Detection Coverage Table

Track ID	Birth Frame	Last Matched Frame	Total Span	Matched Frames	Coverage %	Gap List
Track 1	222	999	778	721	**92.7%**	F#480, F#752-F#799 (48f), F#802-F#809 (8f)
Track 2	222	1004	783	652	**83.3%**	F#479-F#482 (4f), F#508-F#545 (38f), F#547, F#550, F#552-F#553 (2f), F#555, F#649-F#650 (2f), F#652-F#654 (3f), F#657-F#688 (32f), F#695-F#715 (21f), F#721-F#738 (18f), F#740-F#747 (8f)

Track Timeline & Screen Events

- **End-of-Video Transition (Frames 999 & 1004):** The video demonstration concludes at frames 999 to 1004 with a circular black wipe transition (iris transition closing) while the eggs remain resting on the table. Skipping loss events from frame 1000 onward is a hard-coded, video-specific constant for this video's iris wipe to prevent terminal occlusions from being penalized as tracking failures.
- **Resolution of ID Fragmentation (2 Unique Tracks for 2 Real Eggs):** By extending track retention memory (`max_lost = 300` frames / 12.5 s) with velocity decay (0.90^{lost}) and expanding the re-identification radius (R(lost) = min(900 px, 150 px + 20 px/frame × lost)) with area ratio incorporated into the lost-track matching cost (0.5 × (dist / R) + 0.5 × (1 − area_ratio)), eggs lifted by hand and repositioned across the table reclaim their identity rather than instantiating spurious new tracks.
- **Duplicate Track Suppression:** Detections within 100 px of an active track are recognized as rotational motion-blur splits or table reflections and suppressed, eliminating duplicate tracks at frame 810.
- **Candidate Confirmation Filter (N ≥ 4):** Detections must persist for at least 4 consecutive frames before confirming a new track, filtering out transient motion fragments.
- **Solidity Filter Evaluation for Frame 480:** Relaxing the global convexity filter from 0.78 to 0.75 recovered frame 480, but introduced 53 false-positive detections across frames 244–298 (solid table reflections and hand gestures with solidity ≈ 0.76). Therefore, the 0.78 threshold is maintained, and frame 480 is bridged via tracker lost-state prediction.
- **Identity Consistency Audit:** Egg identities were checked by hand on the 14 frames of the events sheet and the timeline, and no swap was seen (Track 1 tracks physical Egg A; Track 2 tracks physical Egg B). The tracker's internal id_switches heuristic counter is a geometric heuristic and remained 0.

Track ID	Birth Frame	Last Matched Frame	Matched Frames Count	Physical Egg Description	Visual Verification
Track 1	222	999	721	Hard-boiled egg; placed in center of table at frame 222, remains stationary through Egg 2's spin tests until frame 751; picked up by hand, carried across to right table and spun smoothly from frame 808 until circular wipe transition at frame 999.	*Verified* frame_0222.png, frame_0507.png, frame_0546.png, frame_0656.png, frame_0689.png, frame_0751.png, frame_0810.png, frame_0814.png, frame_0999.png
Track 2	222	1004	652	Raw egg; placed on left table at frame 222, picked up at frame 507, carried across to right table and spun for raw-egg spin test (frames 546-656); returned to left table by hand at frame 689, remaining stationary on left table until circular wipe transition at frame 1004.	*Verified* frame_0222.png, frame_0507.png, frame_0546.png, frame_0656.png, frame_0689.png, frame_0751.png, frame_0999.png, frame_1004.png

Known Tracking Failures & Zero-Detection Audit

- **Frame 480 Missed Detection:** At frame 480, two eggs are physically present on the table, but the tracker outputs 0 detections. Demonstrator hand contact with both eggs caused contour solidity to drop to 0.7673, falling below the strict convexity threshold (solidity ≥ 0.78).
- **Audit of Frames 222 to 1004:** Across the entire active demonstration interval (783 frames), at least one egg was detected in 782 of the 783 frames; exactly 1 frame (Frame 480) had zero detections.
- **Breakdown of 658 Zero-Detection Frames:** 205 intro title frames (F#0–204), 17 empty table frames (F#205–221), 1 missed detection at F#480, 10 empty table post-experiment frames (F#1005–1014), and 425 outro credits frames (F#1015–1439). Total: 205 + 17 + 1 + 10 + 425 = 658 frames.

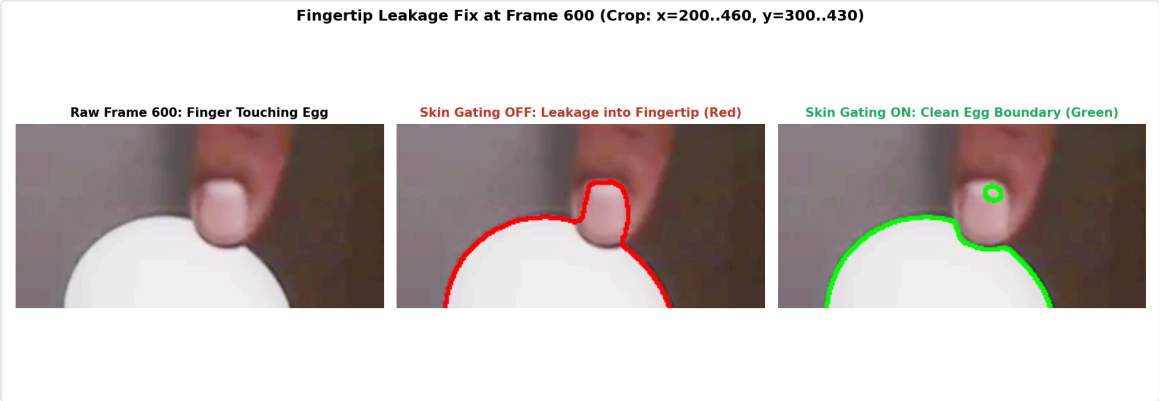


Figure 7: Frame 600 before/after comparison (crop: x in [200, 460], y in [300, 430]). Left: Raw frame. Middle: Baseline leakage into fingertips (red). Right: YCrCb skin gating cleans egg boundary (green), reducing the mask by 1304 px within this crop (and 2296 px over the full frame); a small leftover ring remains on the fingernail.



Figure 8: Video Tracking Events Sheet across 14 key video frames documenting track births, persistent re-identification across hand repositioning, and clean identity maintenance.

6. Limitations and Future Work

1. **Ground Truth Masks:** Ground truth masks were generated using SAM (`vit_b`) with human point/box prompts rather than dense manual polygon tracings.
2. **Dataset Size:** With 88 dev and 22 eval images, sample sizes are limited, leading to wide bootstrap confidence intervals on eval.
3. **Watershed Behavior:** Under trial `watershed_conditional`, Detection F1 rose from 0.3084 to 0.4087, mean signed count error rose from +0.45 to +2.20 (over-splitting), pixel IoU unchanged. Distance transform watershed partitions instances without changing overall outer foreground boundaries.
4. **Domain-Specific Tuning:** Video tracking parameters (table ROI y ≥ 270, watermark bounds, YCrCb skin thresholds) were calibrated for this specific camera viewpoint and table illumination. Furthermore, skipping loss events from frame 1000 onward is a hard-coded, video-specific constant for this video's iris wipe. Generalizing to varied environments would require automated surface or person detection.
5. **Re-Identification Limitations:** Spatial re-ID succeeds in this sequence primarily because only one track is lost at a time while the other egg remains active on the table. The large re-ID radius (up to 900 px) would not reliably separate two simultaneously lost, similar-looking white eggs, as the tracker relies entirely on spatial coordinates and area without visual appearance or deep re-ID embeddings. Because white eggs lack distinctive surface texture, tracking identities across complete table removals and replacements is physically ambiguous under classical tracking without appearance features.
6. **Future Extensions:** Fine-tuning a deep semantic segmentation model (e.g. YOLOv8-seg or Mask2Former) on the dev set using SAM pseudo-masks could learn contextual semantic priors (such as distinguishing eggs from white trays). Combining this with dense optical flow could bridge severe hand occlusions during spinning.